# Week 4: Stars and wheels

Adult guide built from the problem files · Unpiloted.

The answers, hints and routes are in the adult guide, week-04-facilitator.pdf, whose source is guide-src/facilitator.tex. First route = the guide's core problems.

## Problems and routes

| Problem | K–1 | 2–3 | 4–5 | route |
|---|---|---|---|---|
| Hop around a ring of 4 | P1 | | | first |
| Hop around a ring of 5 | P2 | | | first |
| Hop around a ring of 6 | P3 | | | first |
| Start again until every dot has a line | P4 | | | first |
| Which hops draw each picture? | P5 | | | return |
| Hop around a ring of 7: hops 1, 2, 3 | P6 | | | return |
| Hop around a ring of 7: hops 4, 5, 6 | P7 | | | return |
| Hop 2 on rows of pictures | P8 | | | return |
| Which hop changes it back? | P9 | | | return |
| A secret hop | P10 | | | return |
| Hops 1 to 4 on rings of 8 | | P1 | | first |
| Hops 2 to 5 on rings of 10 | | P2 | | first |
| Guess, then draw: rings of 12 | | P3 | | first |
| The wheel: decode five codes at D | | P4 | | first |
| Code a message for your partner | | P5 | | first |
| Guess, then draw: four more rings | | P6 | | return |
| Exactly 3 starts | | P7 | | return |
| Crack two codes with unknown settings | | P8 | | return |
| A second setting that gives the word back | | P9 | | return |
| Hops 1 to 6 on rings of 12: pieces | | | P1 | first |
| Six mixed rings: pieces | | | P2 | first |
| Pieces without drawing | | | P3 | first |
| The wheel: crack four codes | | | P4 | first |
| A message with a secret setting | | | P5 | return |
| Two settings in a row as one | | | P6 | first |
| A rule for the number of pieces | | | P7 | first |
| Rings where every hop makes one piece | | | P8 | return |
| Drawings with the dots erased | | | P9 | return |
| Is coding twice harder to crack? | | | P10 | return |
| Settings that take A through all 26 letters | | | P11 | return |
| Times-table rings: x2 and x3 | | | P12 | return |

**First route.** K–1 P1–P4; Grades 2–3 P1–P5; Grades 4–5 P1–P4, P6–P7. **Return visits.** K–1 P5–P10; Grades 2–3 P6–P9; Grades 4–5 P5, P8–P12.

## Key, one entry per problem

These problem files hold no keys; the week's adult guide has them.